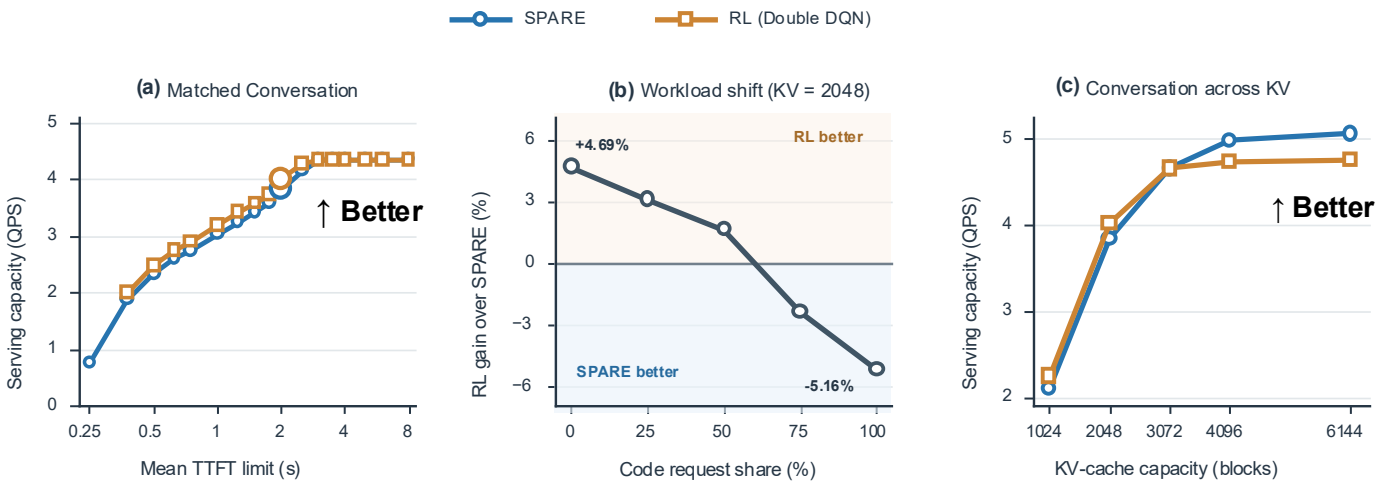

SPARE
RL (Double DQN)
(a) Matched Conversation
Serving capacity (QPS)
↑ Better
Mean TTFT limit (s)
0.25
0.5
1
2
4
8
(b) Workload shift (KV = 2048)
RL gain over SPARE (%)
+4.69%
RL better
SPARE better
-5.16%
Code request share (%)
(c) Conversation across KV
Serving capacity (QPS)
↑ Better
KV-cache capacity (blocks)
1024
2048
3072
4096
6144